Miha E. Habič

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Employment

Proof School
Mathematics faculty *July 2025–present*

Bard College at Simon's Rock
Assistant professor in Mathematics *July 2019–June 2025*

Czech Technical University in Prague
Postdoctoral researcher *September 2018–May 2019*

Charles University
Postdoctoral researcher *August 2017–June 2019*

Hunter College, CUNY
Graduate teaching fellow *2014–2017*

Education

The Graduate Center, CUNY
PhD in Mathematics *2017*

Thesis title: *Joint Laver diamonds and grounded forcing axioms*

Advisor: Joel David Hamkins

University of Ljubljana
MSc in Mathematics *2012*

University of Ljubljana
BSc in Mathematics *2010*

Research interests

Mathematical logic and set theory, particularly large cardinals, their interaction with forcing, and forcing axioms; the structure of forcing extensions of countable models of set theory; computability theory; writing in the Math (and STEM) curriculum.

Publications

Miha E. Habič and Kaethe Minden. Writing in the zones for the reading of proofs in the mathematics classroom: A how-to guide. *Early College Folio*, 3(1):49–58, 2024.

Miha E. Habič, Charles Weng, and Cathy Zhang. More nonamalgamable forcing extensions. In preparation, 2024.

Miha E. Habič and Radek Honzík. Capturing sets of ordinals by normal ultrapowers. *Annals of Pure and Applied Logic*, 174(6):Paper No. 103261, 24, 2023.

Erin Carmody, Victoria Gitman, and Miha E. Habič. A Mitchell-like order for Ramsey and Ramsey-like cardinals. *Fundamenta Mathematicae*, 248(1):1–32, 2020.

Miha E. Habič, Joel David Hamkins, Lukas Daniel Klausner, Jonathan Verner, and Kameryn J. Williams. Set-theoretic blockchains. *Archive for Mathematical Logic*, 58(7-8):965–997, 2019.

Miha E. Habič. Joint diamonds and Laver diamonds. *The Journal of Symbolic Logic*, 84(3):895–928, 2019.

Miha E. Habič. The grounded Martin's axiom. *MLQ. Mathematical Logic Quarterly*, 63(5):437–453, 2017.

Miha E. Habič. *Joint Laver diamonds and grounded forcing axioms*. PhD thesis, The Graduate Center, CUNY, 2017.

Miha E. Habič. Cardinal-recognizing infinite time Turing machines. In *The nature of computation*, volume 7921 of *Lecture Notes in Comput. Sci.*, pages 231–240. Springer, Heidelberg, 2013.

Teaching experience

Proof School:
- *Differential Calculus*
- *Point-Set Topology*
- *Statistics A*
- *Algebra 2A*

Bard College at Simon's Rock:
- MATH 01/02 *Algebra I/II*
- MATH 099 *Algebra Workshop*
- MATH 105 *Math of Games & Puzzles*
- MATH 107 *Logic*
- MATH 109 *Elementary Functions*
- MATH 210 *Calculus I*
- MATH 217 *Intro to Mathematical Proof*
- MATH 220 *Linear Algebra*
- MATH 221 *Vector Calculus*
- MATH 320 *Group Theory*
- MATH 330 *Probability Theory*
- CMPT 320 *Theory of Computation*
- Tutorials in Set Theory and Topics in Cryptography

Hunter College:
- MATH 125 *Precalculus*
- MATH 156 *Introduction to Mathematical Proofs*

Professional service

Mentoring

Undergraduate research lead
Bard College at Simon's Rock *2022*
Supervised two undergraduate research interns (Charles Weng & Catchy Zhang) on summer research project in set theory

Undergraduate theses:

Thesis advisor
Jenny Yang (2024): *Zero-knowledge Proofs*

Committee member
Ariel Rozen (2023): *Ranks of Outer Automorphism Groups of Right-Angled Coxeter Groups*
Maya Saraya (2023): $M_a \times M_u$*: Calculating Composition Technique*
Ethan Xu (2022): *Heat Equation*

Putnam exam coach
Bard College at Simon's Rock *2021–2023*

Scholarly service

Referee/reviewer: Ars Mathematica Contemporanea; Computability; Theoretical Computer Science; Fundamenta Mathematicae; Mathematical Logic Quarterly; Cambridge University Press; Lecture Notes in Computer Science; MathSciNet/Mathematical Reviews; zbMATH

Coorganizer (with V. Gitman)
CUNY Set Theory seminar *2015–2017*

Coorganizer (with K. Minden and J. Williams)
CUNY Student Set Theory seminar *2013–2017*

Campus service

Chair of Code of Conduct hearing panels
Bard College at Simon's Rock *2020–2023*

Member of International Student Support Team
Bard College at Simon's Rock *2019–present*

Math placement coordinator
Bard College at Simon's Rock *2020–present*

Selected conference talks

Some results on ultrapower capturing: Winter School in Abstract Analysis, Hejnice, January 2019

Embedding posets into the set-generic multiverse: Forcing Project Networking Conference, Konstanz, September 2018

Nonamalgamation in the generic multiverse: Novi Sad Conference in Set Theory and General Topology, Novi Sad, July 2018

Surgery and nonamalgability for Cohen reals: Winter School in Abstract Analysis, Hejnice, January 2018

Restricting forcing axioms to ground models: European Set Theory Conference, Budapest, July 2017

A Mitchell-like order for Ramsey cardinals: Joint Mathematics Meetings, Atlanta, January 2017

The grounded Martin's axiom: NY Graduate Student Logic Conference, The Graduate Center, CUNY, May 2016

Joint Laver diamonds: Set Theory Day, The Graduate Center, CUNY, March 2016

Joint Laver diamonds: BEST, San Francisco State University, June 2015

Restricting Martin's axiom to a ccc ground model: ASL Logic Colloquium, Vienna University of Technology, July 2014

Restricting Martin's axiom to a ccc ground model: Joint Mathematics Meetings, Baltimore, January 2014

Cardinal-recognizing infinite time Turing machines: Computability in Europe, Milan, July 2013

Selected seminar talks

Some old and new results on nonamalgamable forcing extensions: CUNY Set Theory seminar, May 2023

Hiding information in generic sequences: Simon's Rock/BHSEC Math seminar, November 2021

Normal ultrapowers with many sets of ordinals: CUNY Set Theory seminar, August 2020

Capturing powersets by normal ultrapowers: CUNY Logic Workshop, October 2019

The generic multiverse, amalgamability, and blockchains: University of Ljubljana, May 2019

Capturing powersets by ultrapowers: Kurt Gödel Research Center, March 2019

The ultrapower capturing property (parts I & II): Charles University Set Theory seminar, January 2019

Surgery and generic coding: CUNY Set Theory seminar, October 2018

Surgery and generic coding: Charles University Set Theory seminar, October 2018

Tukey classes of complete ultrafilters: CUNY Set Theory seminar, May 2018

Amalgamability between Cohen extensions: Charles University Set Theory seminar, March 2018

Joint guessing principles: Charles University Set Theory seminar, November 2017

The grounded Martin's axiom: Charles University Set Theory seminar, September 2017

Bukovský's theorem on forcing extensions: CUNY Set Theory seminar, November 2016

The grounded Martin's axiom: Rutgers University, April 2016

Some guessing principles in set theory: Virginia Commonwealth University, April 2016

The Mitchell order for Ramsey cardinals: CUNY Set Theory seminar, October 2015

Cardinal-recognizing infinite time Turing machines: Virginia Commonwealth University, March 2014
Infinite time Turing machines: University of Ljubljana, December 2012